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| 1                                                                                                                                                                                                                          |  | 2                   |  | 3          |  | 4          |  | 5                                  |  | 6  |  | 7                      |  | 8    |  |
| A                                                                                                                                                                                                                          |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |      |  |
| B                                                                                                                                                                                                                          |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |      |  |
| C                                                                                                                                                                                                                          |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |      |  |
| D                                                                                                                                                                                                                          |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |      |  |
| E                                                                                                                                                                                                                          |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |      |  |
| F                                                                                                                                                                                                                          |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |      |  |
| <div>UniSec</div> <div>Medium Voltage Switchgear</div> <div>PT MULTIMAS NABATI ASAHAN</div> <div>MVD-20 &amp; MVD-21 20kV Boiler Plant</div> <div>SCHEMATIC DIAGRAM</div> <div>SBC-W OUTGOING</div> <div>H03,H09,H10</div> |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |      |  |
|                                                                                                                                                                                                                            |  |                     |  | Date Prep. |  | 15.11.2018 |  | PROJECT TITLE                      |  |    |  | ABB                    |  | EPMV |  |
|                                                                                                                                                                                                                            |  |                     |  | Prepared   |  | YR         |  | MV Switchgear for MNA Serang Plant |  |    |  |                        |  |      |  |
| 00                                                                                                                                                                                                                         |  | Issued for Approval |  | 11.01.19   |  | YR         |  | Checked                            |  | AW |  | PT. ABB Sakti Industri |  |      |  |
| Ind.                                                                                                                                                                                                                       |  | Revision            |  | Date       |  | Name       |  | Approved                           |  | AI |  | Based on: BE318042     |  |      |  |
| 1                                                                                                                                                                                                                          |  | 2                   |  | 3          |  | 4          |  | 5                                  |  | 6  |  | 7                      |  | 8    |  |

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|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
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|                                    | 1                                  | 2                                  | 3        | 4                         | 5                                                  | 6                       | 7                                                   | 8        |    |                        |                    |                                                                                            |          |              |                           |               |
| A                                  | SHEET INDEX                        |                                    |          |                           |                                                    |                         |                                                     |          | A  |                        |                    |                                                                                            |          |              |                           |               |
|                                    | PAGE                               |                                    |          |                           | PAGE NAME                                          |                         |                                                     | REVISION |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/1  |                                    |          |                           | COVER SHEET                                        |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/2  |                                    |          |                           | Index of Sheet                                     |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/3  |                                    |          |                           | REFERENCE DESIGNATIONS                             |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/4  |                                    |          |                           | REFERENCE DESIGNATIONS                             |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/5  |                                    |          |                           | REFERENCE DESIGNATIONS                             |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/6  |                                    |          |                           | REFERENCE DESIGNATIONS                             |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/7  |                                    |          |                           | DISTRIBUTION OF AUXILIARY CIRCUITS                 |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/8  |                                    |          |                           | AUXILIARY SUPPLY CIRCUIT                           |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    | ==MVD-20_MVD-21=H03,H09,H10&305/9  |          |                           |                                                    | CIRCUIT BREAKER CONTROL |                                                     |          | 00 |                        |                    |                                                                                            |          |              |                           |               |
| ==MVD-20_MVD-21=H03,H09,H10&305/10 |                                    |                                    |          | AUXILIARY MOTOR & RELAY   |                                                    |                         | 00                                                  |          |    |                        |                    |                                                                                            |          |              |                           |               |
| ==MVD-20_MVD-21=H03,H09,H10&305/11 |                                    |                                    |          | DIGITAL INPUT/OUPUT RELAY |                                                    |                         | 00                                                  |          |    |                        |                    |                                                                                            |          |              |                           |               |
| ==MVD-20_MVD-21=H03,H09,H10&305/12 |                                    |                                    |          | SPARE CONTACT             |                                                    |                         | 00                                                  |          |    |                        |                    |                                                                                            |          |              |                           |               |
| ==MVD-20_MVD-21=H03,H09,H10&305/13 |                                    |                                    |          | CURRENT TRANSFOMER        |                                                    |                         | 00                                                  |          |    |                        |                    |                                                                                            |          |              |                           |               |
| ==MVD-20_MVD-21=H03,H09,H10&305/14 |                                    |                                    |          | VOLTAGE TRANSFORMER       |                                                    |                         | 00                                                  |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    | ==MVD-20_MVD-21=H03,H09,H10&305/15 |          |                           |                                                    | LV DOOR LAYOUT          |                                                     |          | 00 |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/16 |                                    |          |                           | LV BOX LAYOUT                                      |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/17 |                                    |          |                           | PART LIST                                          |                         |                                                     | 00       |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/18 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDA  |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/19 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDB1 |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/20 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDC1 |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/21 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDC2 |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/22 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDI  |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/23 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDI2 |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/24 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDI3 |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/25 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDI6 |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/26 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDV  |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    | ==MVD-20_MVD-21=H03,H09,H10&305/27 |                                    |          |                           | Terminal diagram ==MVD-20_MVD-21=H03,H09,H10+-XDX  |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           |                                                    |                         |                                                     |          |    |                        |                    |                                                                                            |          |              |                           |               |
|                                    |                                    |                                    |          |                           | Date Prep.                                         | 15.11.2018              | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |          |    | ABB                    | EPMV               | Index of Sheet<br>MVD-20 & MVD-21 20kV Boiler Plant<br>SCHEMATIC DIAGRAM<br>SBC-W OUTGOING |          | Scale        | NTS                       | = H03,H09,H10 |
|                                    |                                    |                                    |          |                           | Prepared                                           | YR                      |                                                     |          |    |                        |                    |                                                                                            |          | Customer     | PT Multimas Nabati Asahan | & 305         |
|                                    | 00                                 | Issued for Approval                | 11.01.19 | YR                        | Checked                                            | AW                      |                                                     |          |    |                        |                    |                                                                                            |          | Drawing Nbr. | 18.042-MNA-305.03         | Sh. 2         |
|                                    | Ind.                               | Revision                           | Date     | Name                      | Approved                                           | AI                      | Based on:                                           | BE318042 |    | PT. ABB Sakti Industri | Customer Doc. Nbr. |                                                                                            | Cont. 27 |              |                           |               |
|                                    | 1                                  | 2                                  | 3        | 4                         | 5                                                  | 6                       | 7                                                   | 8        |    |                        |                    |                                                                                            |          |              |                           |               |

(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)

|      |               |                     |                                                                                                                                                     |      |                                   |                                                                                                                                     |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|------|---------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------|--------------------|--------------|---------------------------|-------|--|--|---|--|--|---|--|--|
| B    | DESIGNATION   |                     | DESCRIPTION                                                                                                                                         |      | -BGB8...-BGB10                    | POSITION SWITCHES OF CIRCUIT-BREAKER                                                                                                | B                                  |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -AA           |                     | MULTIFUNCTION UNIT (CENTRAL UNIT)                                                                                                                   |      | -BGB11                            | POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING                                                                     |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAD1,-BAD4   |                     | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1                                                                                                      |      | -BGB12                            | PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION                                                                        |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAD2,-BAD5   |                     | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2                                                                                                      |      | -BGB13                            | PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION                                                                      |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAD3,-BAD6   |                     | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3                                                                                                      |      | -BGD                              | POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.                                                                                |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAR          |                     | VOLTAGE PROTECTION RELAY                                                                                                                            |      | -BGD1                             | POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR                                                                                 |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAS1         |                     | VOLTAGE SENSOR LOCATED ON PHASE L1                                                                                                                  |      | -BGD2                             | POSITION SWITCH OF CABLE COMPARTMENT DOOR                                                                                           |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAS2         |                     | VOLTAGE SENSOR LOCATED ON PHASE L2                                                                                                                  |      | -BGE1                             | POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION                                                                  |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAS3         |                     | VOLTAGE SENSOR LOCATED ON PHASE L3                                                                                                                  |      | -BGE2                             | POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION                                                                |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAT1         |                     | VOLTAGE TRANSFORMER LOCATED ON PHASE L1                                                                                                             |      | -BGE3                             | POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)              |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
| C    | -BAT2         |                     | VOLTAGE TRANSFORMER LOCATED ON PHASE L2                                                                                                             |      | -BGE4                             | POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR                                                   | C                                  |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAT3         |                     | VOLTAGE TRANSFORMER LOCATED ON PHASE L3                                                                                                             |      | -BGE5                             | POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR                                         |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAT4         |                     | VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1                                                                                       |      | -BGE7                             | POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION                                                                  |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAT5         |                     | VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2                                                                                       |      | -BGE8                             | POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS                                          |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BAT6         |                     | VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3                                                                                       |      | -BGE11                            | POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION                                                        |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCD          |                     | DIFFERENTIAL PROTECTIVE RELAY                                                                                                                       |      | -BGE12                            | POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION                                                      |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCF          |                     | FEEDER PROTECTION RELAY                                                                                                                             |      | -BGE13                            | POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)    |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCG          |                     | GENERATOR PROTECTION RELAY                                                                                                                          |      | -BGE14, -BGE15                    | POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCM          |                     | MOTOR PROTECTION RELAY                                                                                                                              |      | -BGF                              | POSITION SWITCHES OF MEDIUM VOLTAGE FUSES                                                                                           |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCN          |                     | NEUTRAL (RESIDUAL) CURRENT TRANSFORMER                                                                                                              |      | -BG11                             | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION                                                                     |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
| D    | -BCP          |                     | TRANSFORMER PROTECTION RELAY                                                                                                                        |      | -BG12                             | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION                                                                   | D                                  |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCR          |                     | CURRENT PROTECTION RELAY                                                                                                                            |      | -BG13                             | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)                 |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCS1         |                     | CURRENT SENSOR LOCATED ON PHASE L1                                                                                                                  |      | -BG14                             | POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR                                                     |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCS2         |                     | CURRENT SENSOR LOCATED ON PHASE L2                                                                                                                  |      | -BG15                             | POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR                                                     |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCS3         |                     | CURRENT SENSOR LOCATED ON PHASE L3                                                                                                                  |      | -BG16                             | POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION                             |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCT1         |                     | CURRENT TRANSFORMER LOCATED ON PHASE L1                                                                                                             |      | -BG17                             | POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION                                        |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCT2         |                     | CURRENT TRANSFORMER LOCATED ON PHASE L2                                                                                                             |      | -BG18                             | POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS                                                                                       |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCT3         |                     | CURRENT TRANSFORMER LOCATED ON PHASE L3                                                                                                             |      | -BG19                             | POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR                                               |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCT4,-BCT7   |                     | CURRENT TRANSFORMER LOCATED ON PHASE L1                                                                                                             |      | -BG111                            | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION                                                                    |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCT5,-BCT8   |                     | CURRENT TRANSFORMER LOCATED ON PHASE L2                                                                                                             |      | -BG112                            | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION                                                                  |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
| E    | -BCT6,-BCT9   |                     | CURRENT TRANSFORMER LOCATED ON PHASE L3                                                                                                             |      | -BG113                            | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)                | E                                  |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BCZ          |                     | DISTANCE PROTECTION RELAY                                                                                                                           |      | -BG114, -BG115                    | POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR                                                     |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BEF          |                     | FREQUENCY PROTECTION RELAY                                                                                                                          |      | -BG121                            | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION                                                                    |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BER          |                     | SUPERVISION RELAYS                                                                                                                                  |      | -BG122                            | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION                                                                  |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BES          |                     | SYNCHRONIZING RELAY                                                                                                                                 |      | -BG123                            | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)                |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BET          |                     | THERMAL PROTECTION RELAY                                                                                                                            |      | -BG124, -BG125                    | POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR                                                     |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BGA1         |                     | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT                                                                     |      | REFERENCE DESIGNATIONS            | Scale                                                                                                                               |                                    | NTS                       | = H03,H09,H10                                                                                                                                                               |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BGA2         |                     | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM) |      | MVD-20 & MVD-21 20kV Boiler Plant | Customer                                                                                                                            |                                    | PT Multimas Nabati Asahan | & 305                                                                                                                                                                       |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BGA3         |                     | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT                                                                               |      | SCHEMATIC DIAGRAM                 | Drawing Nbr.                                                                                                                        |                                    | 18.042-MNA-305.03         | Sh. 3                                                                                                                                                                       |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BGA4         |                     | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)                                                  |      | SBC-W OUTGOING                    | Customer Doc. Nbr.                                                                                                                  |                                    |                           | Cont. 27                                                                                                                                                                    |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
| F    | -BGB1...-BGB3 |                     | POSITION SWITCHES OF CIRCUIT-BREAKER                                                                                                                |      |                                   |                                                                                                                                     |                                    |                           | F                                                                                                                                                                           |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      | -BGB4         |                     | POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS                                                      |      |                                   |                                                                                                                                     |                                    |                           |                                                                                                                                                                             |                        |                                   |                    |              |                           |       |  |  |   |  |  |   |  |  |
|      |               |                     |                                                                                                                                                     |      | Date Prep.                        | 15.11.2018                                                                                                                          | PROJECT TITLE                      |                           |   | REFERENCE DESIGNATIONS |                                   | Scale              | NTS          | = H03,H09,H10             |       |  |  |   |  |  |   |  |  |
|      |               |                     |                                                                                                                                                     |      | Prepared                          | YR                                                                                                                                  | MV Switchgear for MNA Serang Plant |                           |                                                                                                                                                                             | PT. ABB Sakti Industri | MVD-20 & MVD-21 20kV Boiler Plant |                    | Customer     | PT Multimas Nabati Asahan | & 305 |  |  |   |  |  |   |  |  |
| 00   |               | Issued for Approval | 11.01.19                                                                                                                                            | YR   | Checked                           | AW                                                                                                                                  |                                    |                           |                                                                                                                                                                             |                        | SCHEMATIC DIAGRAM                 |                    | Drawing Nbr. | 18.042-MNA-305.03         | Sh. 3 |  |  |   |  |  |   |  |  |
| Ind. |               | Revision            | Date                                                                                                                                                | Name | Approved                          | AI                                                                                                                                  | Based on: BE318042                 |                           |                                                                                                                                                                             | SBC-W OUTGOING         |                                   | Customer Doc. Nbr. |              | Cont. 27                  |       |  |  |   |  |  |   |  |  |
| 1    |               |                     | 2                                                                                                                                                   |      |                                   | 3                                                                                                                                   |                                    |                           | 4                                                                                                                                                                           |                        |                                   | 5                  |              |                           | 6     |  |  | 7 |  |  | 8 |  |  |



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|------|----------|----------------------------------------------------------------------------------------------------------------|----------|----------|------------|--------------------|-----------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------|---|--------------------|---------------------------|--|---------------|
| A    | -MAE     | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE                                                         |          |          |            |                    |                                                     | -QBH                   | MANUAL CIRCUIT-BREAKERS                                                                                      |      |                                                             |   |                    |                           |  |               |
|      | -MAE1    | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2                                               |          |          |            |                    |                                                     | -QBL                   | LINKS                                                                                                        |      |                                                             |   |                    |                           |  |               |
|      | -MAE8    | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS         |          |          |            |                    |                                                     | -QBM                   | MINIATURE SWITCH-DISCONNECTORS                                                                               |      |                                                             |   |                    |                           |  |               |
|      | -MAS     | MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING                                                                     |          |          |            |                    |                                                     | -QBS                   | SWITCH-DISCONNECTORS                                                                                         |      |                                                             |   |                    |                           |  |               |
|      | -MAT     | MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT                                                         |          |          |            |                    |                                                     | -QBT                   | TRUCK                                                                                                        |      |                                                             |   |                    |                           |  |               |
|      | -MBC     | CLOSING RELEASE OF CIRCUIT-BREAKER                                                                             |          |          |            |                    |                                                     | -QCA                   | EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING                                                           |      |                                                             |   |                    |                           |  |               |
|      | -MBC4    | CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS                                                                    |          |          |            |                    |                                                     | -QCE                   | EARTHING SWITCH                                                                                              |      |                                                             |   |                    |                           |  |               |
|      | -MBO1    | FIRST OPENING RELEASE OF CIRCUIT-BREAKER                                                                       |          |          |            |                    |                                                     | -QCE1                  | EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)                 |      |                                                             |   |                    |                           |  |               |
|      | -MBO2    | SECOND OPENING RELEASE OF CIRCUIT-BREAKER                                                                      |          |          |            |                    |                                                     | -QCE2                  | EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)                                                       |      |                                                             |   |                    |                           |  |               |
|      | -MBO3    | OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER |          |          |            |                    |                                                     | -QQ                    | CLUTCHES                                                                                                     |      |                                                             |   |                    |                           |  |               |
| B    | -MBO4    | OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS                                                                    |          |          |            |                    |                                                     | -RAA                   | ANTIFERRORESONANCE RESISTOR                                                                                  |      |                                                             |   |                    |                           |  |               |
|      | -MBU     | UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER                                                                        |          |          |            |                    |                                                     | -RAD                   | DIODES                                                                                                       |      |                                                             |   |                    |                           |  |               |
|      | -PFB     | BLUE SIGNAL LAMPS                                                                                              |          |          |            |                    |                                                     | -RAI                   | INDUCTORS                                                                                                    |      |                                                             |   |                    |                           |  |               |
|      | -PFF     | FLAG RELAYS                                                                                                    |          |          |            |                    |                                                     | -RAR                   | RESISTORS                                                                                                    |      |                                                             |   |                    |                           |  |               |
|      | -PFG     | GREEN SIGNAL LAMPS                                                                                             |          |          |            |                    |                                                     | -RF                    | FILTERS                                                                                                      |      |                                                             |   |                    |                           |  |               |
| C    | -PFR     | RED SIGNAL LAMPS                                                                                               |          |          |            |                    |                                                     | -RLE1                  | ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING                                                    |      |                                                             |   |                    |                           |  |               |
|      | -PFS     | SHORT CIRCUIT INDICATORS                                                                                       |          |          |            |                    |                                                     | -RLE2                  | ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT                                                       |      |                                                             |   |                    |                           |  |               |
|      | -PFV     | VOLTAGE INDICATORS                                                                                             |          |          |            |                    |                                                     | -RLE3                  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE           |      |                                                             |   |                    |                           |  |               |
|      | -PFV1    | VOLTAGE INDICATORS ON FEEDER SIDE                                                                              |          |          |            |                    |                                                     | -RLE4                  | ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION                                                 |      |                                                             |   |                    |                           |  |               |
|      | -PFV2    | VOLTAGE INDICATORS ON BUSBAR SIDE                                                                              |          |          |            |                    |                                                     | -RLE5                  | ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD                                             |      |                                                             |   |                    |                           |  |               |
|      | -PFW     | WHITE SIGNAL LAMPS                                                                                             |          |          |            |                    |                                                     | -RLE6                  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1             |      |                                                             |   |                    |                           |  |               |
|      | -PFX     | CROSS INDICATORS, ELECTROMECHANICAL INDICATORS                                                                 |          |          |            |                    |                                                     | -RLE7                  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2             |      |                                                             |   |                    |                           |  |               |
|      | -PFY     | YELLOW SIGNAL LAMPS                                                                                            |          |          |            |                    |                                                     | -RLE8                  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 |      |                                                             |   |                    |                           |  |               |
|      | -PGA     | AMMETERS                                                                                                       |          |          |            |                    |                                                     | -RLE9                  | ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS                                      |      |                                                             |   |                    |                           |  |               |
|      | -PGC     | COUNTERS                                                                                                       |          |          |            |                    |                                                     | -RLE11                 | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4                     |      |                                                             |   |                    |                           |  |               |
| D    | -PGF     | FREQUENCYMETERS                                                                                                |          |          |            |                    |                                                     | -RLE12                 | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5                     |      |                                                             |   |                    |                           |  |               |
|      | -PGH     | HOURMETERS                                                                                                     |          |          |            |                    |                                                     | -SFA                   | AMMETRIC SWITCHES                                                                                            |      |                                                             |   |                    |                           |  |               |
|      | -PGI     | PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE                                                           |          |          |            |                    |                                                     | -SFC                   | CONTROL SWITCHES, CLOSING PUSH-BUTTONS                                                                       |      |                                                             |   |                    |                           |  |               |
|      | -PGI1    | HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)                          |          |          |            |                    |                                                     | -SFL                   | LOCKING CONTACTS                                                                                             |      |                                                             |   |                    |                           |  |               |
|      | -PGI2    | HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)                              |          |          |            |                    |                                                     | -SFO                   | OPENING PUSH-BUTTONS                                                                                         |      |                                                             |   |                    |                           |  |               |
|      | -PGJ     | ACTIVE ENERGY METERS                                                                                           |          |          |            |                    |                                                     | -SFR                   | RESET PUSH-BUTTONS                                                                                           |      |                                                             |   |                    |                           |  |               |
|      | -PGK     | REACTIVE ENERGY METERS                                                                                         |          |          |            |                    |                                                     | -SFS                   | SELECTOR SWITCHES                                                                                            |      |                                                             |   |                    |                           |  |               |
|      | -PGM     | MULTIFUNCTION INDICATORS                                                                                       |          |          |            |                    |                                                     | -SFT                   | TEST PUSH-BUTTONS                                                                                            |      |                                                             |   |                    |                           |  |               |
|      | -PGP     | POWER-FACTOR METERS                                                                                            |          |          |            |                    |                                                     | -SFU                   | UNLOCKING PUSH-BUTTONS                                                                                       |      |                                                             |   |                    |                           |  |               |
|      | -PGQ     | VARMETERS                                                                                                      |          |          |            |                    |                                                     | -SFV                   | VOLTMETRIC SWITCHES                                                                                          |      |                                                             |   |                    |                           |  |               |
| E    | -PGS     | SYNCHRONOSCOPES                                                                                                |          |          |            |                    |                                                     | -TA                    | POWER TRANSFORMERS                                                                                           |      |                                                             |   |                    |                           |  |               |
|      | -PGV     | VOLTMETERS                                                                                                     |          |          |            |                    |                                                     | -TB                    | CONVERTER                                                                                                    |      |                                                             |   |                    |                           |  |               |
|      | -PGW     | WATTMETERS                                                                                                     |          |          |            |                    |                                                     | -TB1                   | RECTIFIER FOR FIRST OPENING RELEASE                                                                          |      |                                                             |   |                    |                           |  |               |
|      | -PJ      | ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)                                                                      |          |          |            |                    |                                                     | -TB2                   | RECTIFIER FOR SECOND OPENING RELEASE                                                                         |      |                                                             |   |                    |                           |  |               |
|      | -QAB     | CIRCUIT-BREAKERS                                                                                               |          |          |            |                    |                                                     | -TB3                   | RECTIFIER FOR CLOSING RELEASE                                                                                |      |                                                             |   |                    |                           |  |               |
|      | -QAC     | CONTACTORS (FOR POWER)                                                                                         |          |          |            |                    |                                                     | -TB4                   | RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING                                      |      |                                                             |   |                    |                           |  |               |
|      | -QBD     | DISCONNECTORS                                                                                                  |          |          |            |                    |                                                     | -TB5                   | RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT                                         |      |                                                             |   |                    |                           |  |               |
|      | -QBD1    | FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                              |          |          |            |                    |                                                     | -TB6                   | RECTIFIER FOR UNDERVOLTAGE RELEASE                                                                           |      |                                                             |   |                    |                           |  |               |
|      | -QBD2    | REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                               |          |          |            |                    |                                                     | -TB7                   | ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER                                               |      |                                                             |   |                    |                           |  |               |
|      | -QBD3    | FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                                    |          |          |            |                    |                                                     |                        |                                                                                                              |      |                                                             |   |                    |                           |  |               |
| F    | -QBD4    | TWO-WAY DISCONNECTOR                                                                                           |          |          |            |                    |                                                     |                        |                                                                                                              |      |                                                             |   |                    |                           |  |               |
|      | -QBD5    | TWO-WAY DISCONNECTOR ON BUSBAR RISER                                                                           |          |          |            |                    |                                                     |                        |                                                                                                              |      |                                                             |   |                    |                           |  |               |
|      |          |                                                                                                                |          |          | Date Prep. | 15.11.2018         | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |                        | ABB                                                                                                          | EPMV | REFERENCE DESIGNATIONS<br>MVD-20 & MVD-21 20kV Boiler Plant |   | Scale              | NTS                       |  | = H03,H09,H10 |
|      |          |                                                                                                                |          |          | Prepared   | YR                 |                                                     |                        |                                                                                                              |      |                                                             |   | Customer           | PT Multimas Nabati Asahan |  | & 305         |
|      | 00       | Issued for Approval                                                                                            | 11.01.19 | YR       | Checked    | AW                 |                                                     |                        |                                                                                                              |      |                                                             |   | Drawing Nbr.       | 18.042-MNA-305.03         |  | Sh. 5         |
| Ind. | Revision | Date                                                                                                           | Name     | Approved | AI         | Based on: BE318042 |                                                     | PT. ABB Sakti Industri |                                                                                                              |      | SCHEMATIC DIAGRAM<br>SBC-W OUTGOING                         |   | Customer Doc. Nbr. |                           |  | Cont. 27      |
| 1    |          | 2                                                                                                              |          | 3        |            | 4                  |                                                     | 5                      |                                                                                                              | 6    |                                                             | 7 |                    | 8                         |  |               |

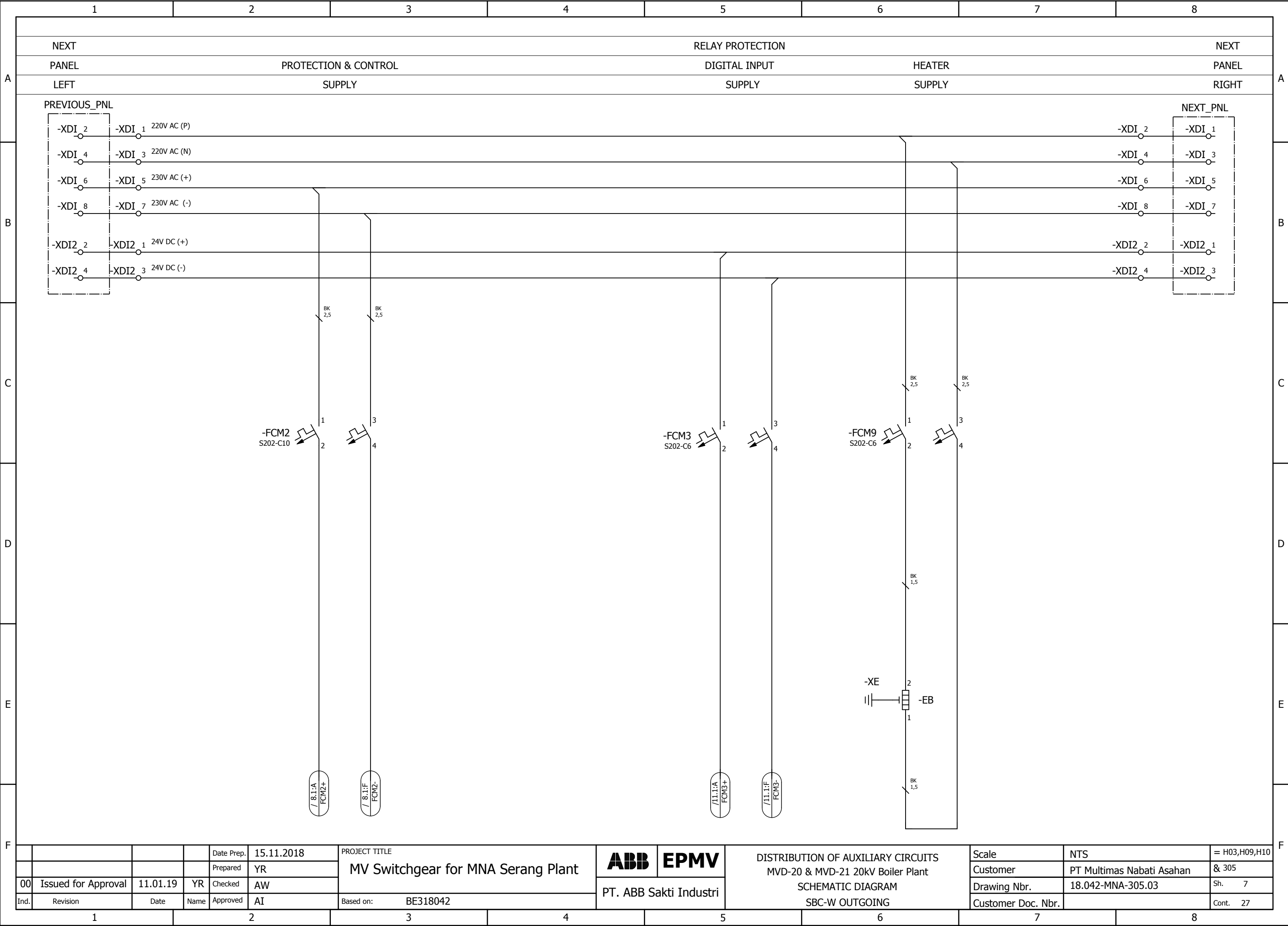
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|-----------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------|
| A               | -TFA                                                                                               | ACTIVE POWER TRANSDUCERS                                           | -XDI1                                               | TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS            | A                                                                  |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFC                                                                                               | CURRENT TRANSDUCERS                                                |                                                     | -XDI2                                                              |                                                                    | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFF                                                                                               | FREQUENCY TRANSDUCERS                                              |                                                     |                                                                    |                                                                    | -XDI3                                                                                         | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFJ                                                                                               | ACTIVE ENERGY TRANSDUCERS                                          |                                                     |                                                                    |                                                                    |                                                                                               | -XDI4                                                                                      | TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS    |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFK                                                                                               | REACTIVE ENERGY TRANSDUCERS                                        |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            | -XDI6                                                             | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFM                                                                                               | MULTIFUNCTION TRANSDUCERS                                          |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFP                                                                                               | POWER-FACTOR TRANSDUCERS                                           |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFQ                                                                                               | REACTIVE POWER TRANSDUCERS                                         |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFS                                                                                               | SIGNAL CONVERTERS                                                  |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFV                                                                                               | VOLTAGE TRANSDUCERS                                                |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| B               | -WA                                                                                                | BUSBARS                                                            | -XDI7                                               |                                                                    | TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS                      |                                                                                               |                                                                                            |                                                                   | B                                                                                  |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WBC                                                                                               | POWER CABLES                                                       |                                                     | -XDI8                                                              | TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS            |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WE                                                                                                | EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.                   |                                                     |                                                                    | -XDI9                                                              | TERMINAL BLOCK FOR SPECIAL USE                                                                |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WGC                                                                                               | CONTROL CABLES                                                     |                                                     |                                                                    |                                                                    | -XDJ                                                                                          | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD                                |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WGS                                                                                               | COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)                      |                                                     |                                                                    |                                                                    |                                                                                               | -XDJ1                                                                                      | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WH                                                                                                | OPTICAL FIBRES                                                     |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            | -XDM                                                              |                                                                                    | SEALABLE TERMINAL BLOCK FOR MEASUREMENT                 |                                                                     |                                                                     |                                                                        |
|                 | -XDA                                                                                               | TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS                |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    | -XDS                                                    | CURRENT SOCKETS                                                     |                                                                     |                                                                        |
|                 | -XDA1                                                                                              | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1 |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         | -XDT                                                                | TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB |                                                                        |
|                 | -XDA2                                                                                              | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2 |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     | -XDU                                                                | CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK |
|                 | -XDA3                                                                                              | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3 |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     | -XDU1                                                                  |
| -XDA5           | CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER                                   | -XDV                                                               | TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB            | CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS |                                                                    | -XDV1                                               | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1 |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB1           | CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)           |                                                                    |                                                     | -XDV2                                                              | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2 |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB2           | CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER                                     |                                                                    |                                                     |                                                                    | -XDV3                                                              | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3                            |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB10...-XDB89 | CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS                                                    |                                                                    |                                                     |                                                                    |                                                                    | -XDV4                                                                                         | CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR                                      |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB7           | CONNECTOR FOR TRUCK POSITION CONTACTS                                                              |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               | -XDX                                                                                       | SUPPORT TERMINAL BLOCK                                            |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB91          | CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            | -XE1                                                              | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE                             |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB92          | CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   | -XE2                                                                               | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE |                                                                     |                                                                     |                                                                        |
| -XDB93          | CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR                                             |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    | -XE5                                                    | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE |                                                                     |                                                                        |
| -XDB95          | CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET                                           |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         | -XE6                                                                | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE       |                                                                        |
| -XDC            | CUSTOMER TERMINAL BLOCK                                                                            | -XE7                                                               |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE              |                                                                        |
| -XDC1           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER                                  |                                                                    | -XE8                                                |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE      |                                                                        |
| -XDC2           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR                              |                                                                    |                                                     | -XE9                                                               |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     | EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT      |                                                                        |
| -XDC3           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT          |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDC4           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS                             |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDC5           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER                        |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDE            | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE                                      |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDE1           | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE                                       |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDE2           | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                            |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDE3           | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                             |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDF            | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS                                |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDF1           | CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR                                            |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDF2           | CONNECTOR FOR REAR VENTILATOR CIRCUITS                                                             |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDF3           | CONNECTOR FOR FRONT VENTILATOR CIRCUITS                                                            |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDH            | TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)   |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDI            | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)                                     |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |

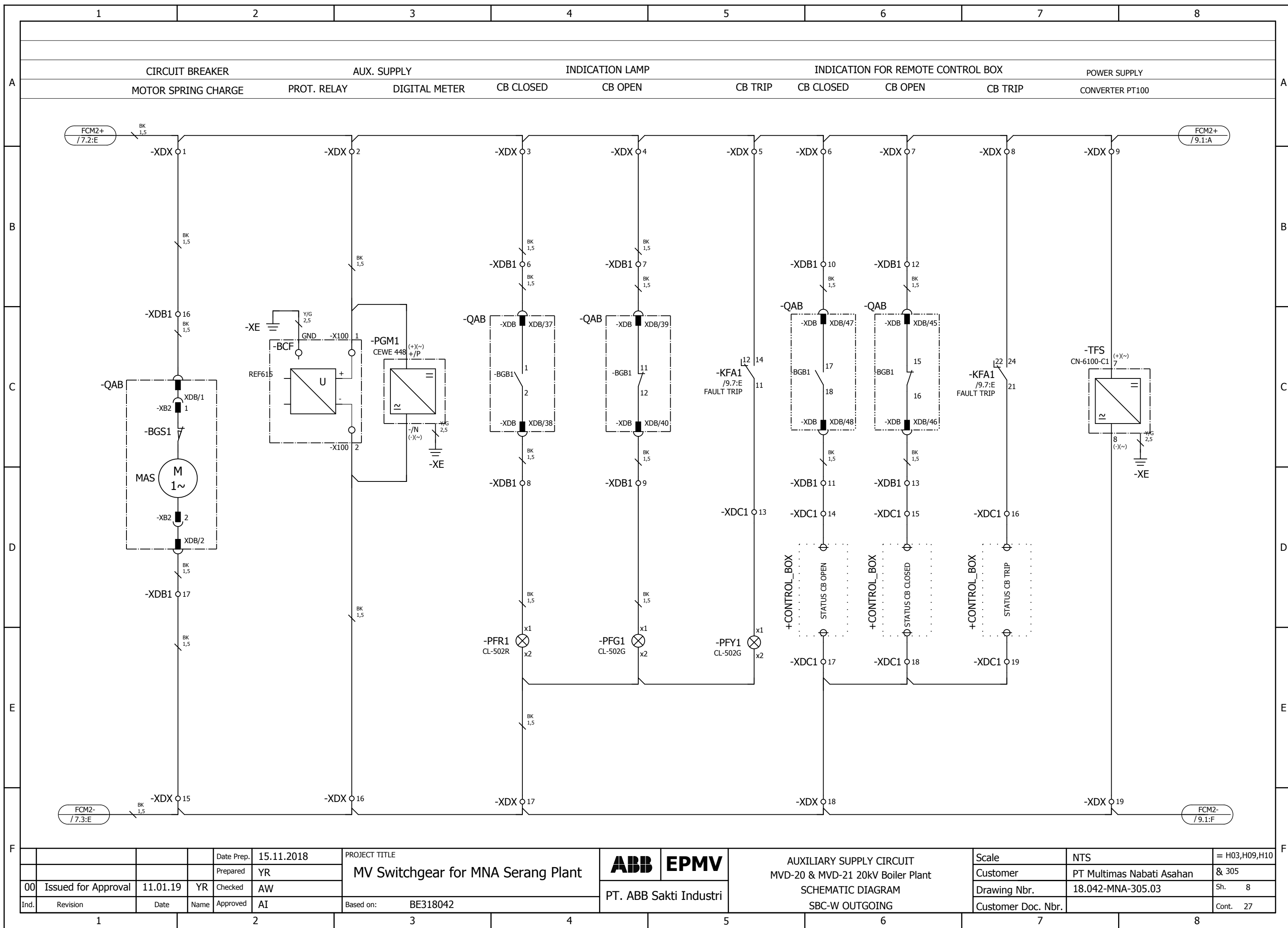
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|------|---------------------|----------|------|------------|------------|-----------------------------------------------------|--|------------------------|------|-------------------------------------------------------------|--|--------------------|---------------------------|---------------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |  | ABB                    | EPMV | REFERENCE DESIGNATIONS<br>MVD-20 & MVD-21 20kV Boiler Plant |  | Scale              | NTS                       | = H03,H09,H10 |
|      |                     |          |      | Prepared   | YR         |                                                     |  |                        |      |                                                             |  | Customer           | PT Multimas Nabati Asahan | & 305         |
| 00   | Issued for Approval | 11.01.19 | YR   | Checked    | AW         |                                                     |  |                        |      | SCHEMATIC DIAGRAM                                           |  | Drawing Nbr.       | 18.042-MNA-305.03         | Sh.           |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on: BE318042                                  |  | PT. ABB Sakti Industri |      | SBC-W OUTGOING                                              |  | Customer Doc. Nbr. |                           | Cont. 27      |

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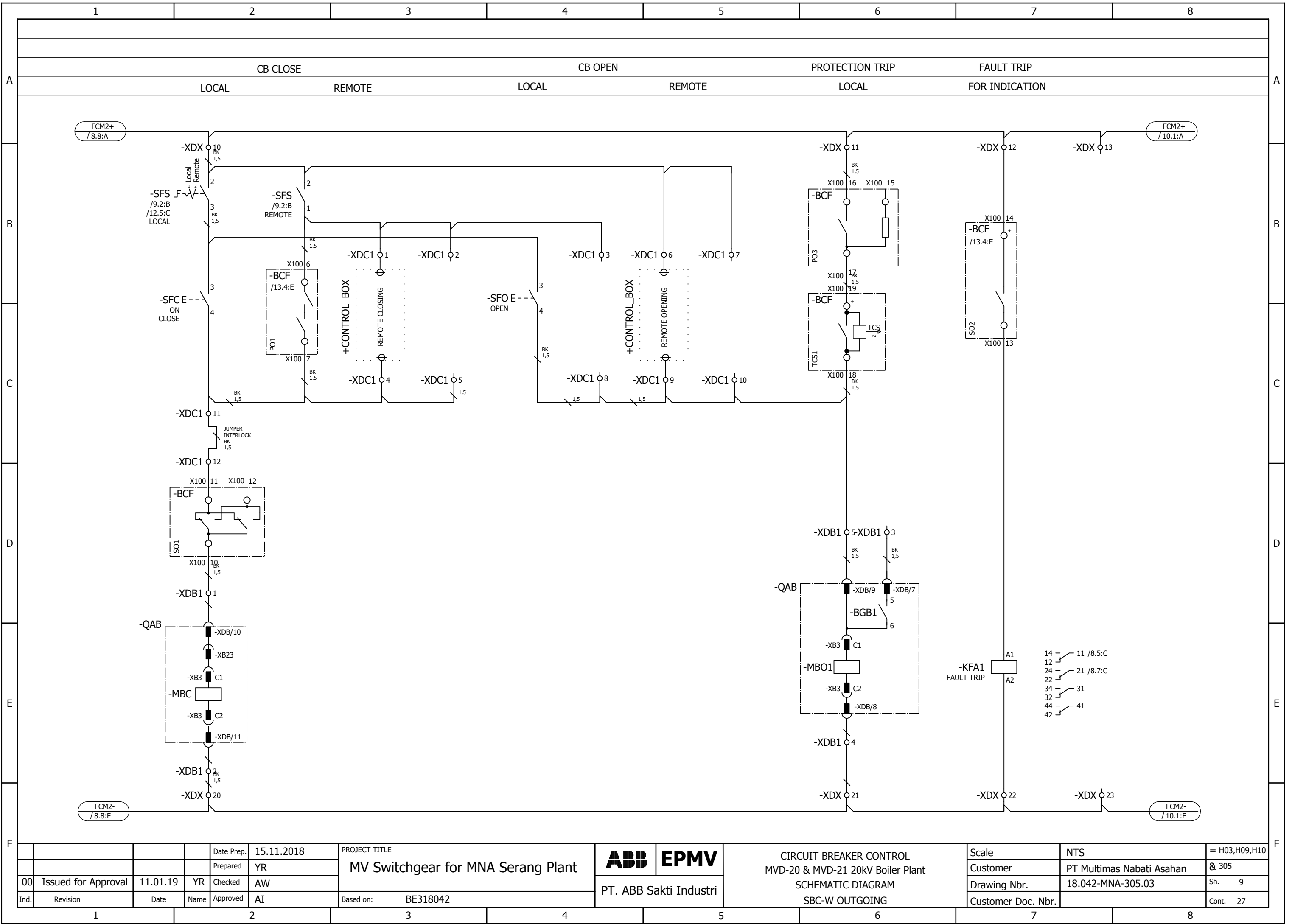
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|      |                     |          |      | Date Prep. | 15.11.2018 |
|      |                     |          |      | Prepared   | YR         |
| 00   | Issued for Approval | 11.01.19 | YR   | Checked    | AW         |
| Ind. | Revision            | Date     | Name | Approved   | AI         |

|                                    |          |
|------------------------------------|----------|
| PROJECT TITLE                      |          |
| MV Switchgear for MNA Serang Plant |          |
| Based on:                          | BE318042 |

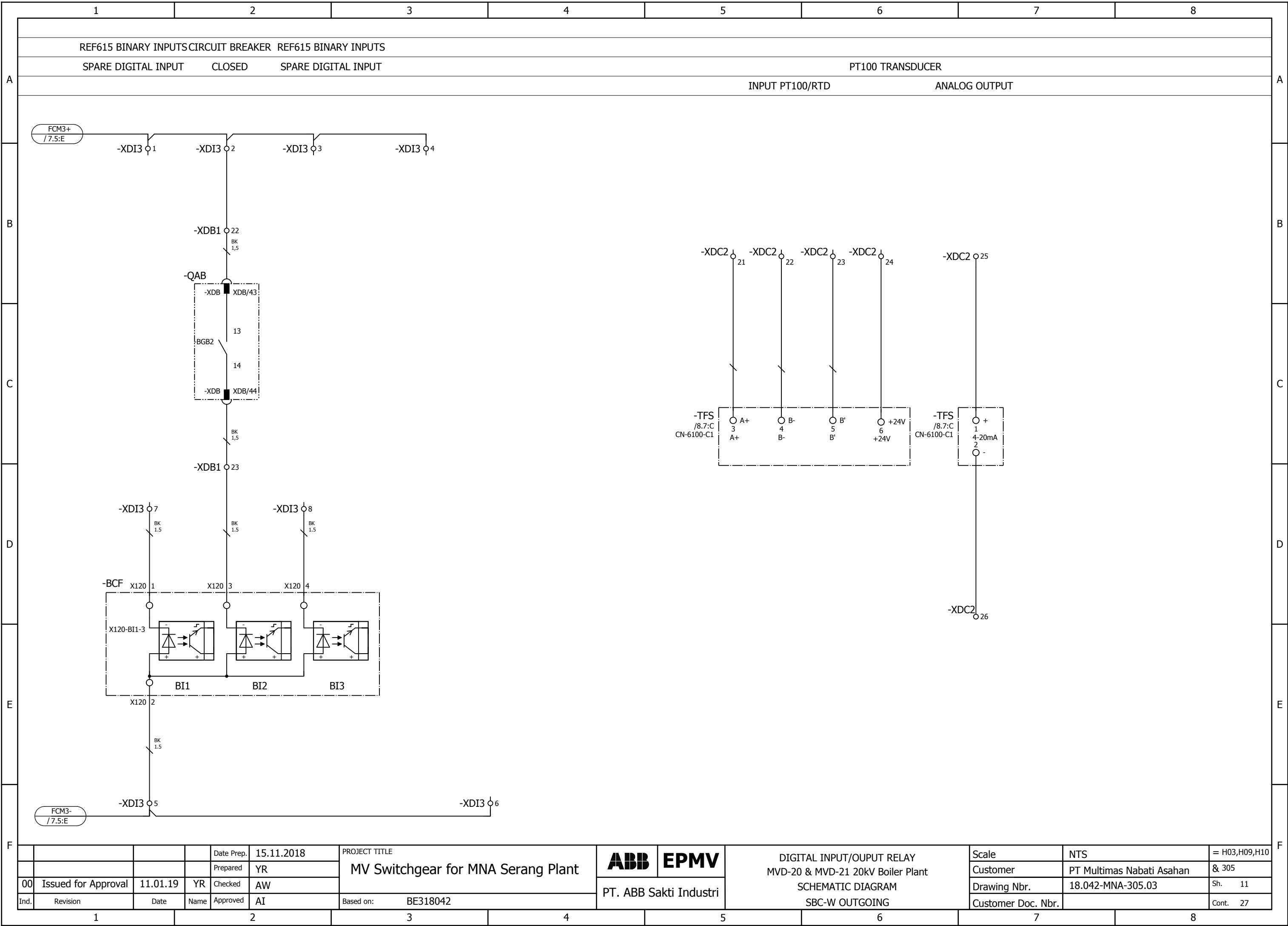
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| <b>ABB</b>             | <b>EPMV</b> |
| PT. ABB Sakti Industri |             |

|                                   |
|-----------------------------------|
| CIRCUIT BREAKER CONTROL           |
| MVD-20 & MVD-21 20kV Boiler Plant |
| SCHEMATIC DIAGRAM                 |
| SBC-W OUTGOING                    |

|                    |                           |               |
|--------------------|---------------------------|---------------|
| Scale              | NTS                       | = H03,H09,H10 |
| Customer           | PT Multimas Nabati Asahan | & 305         |
| Drawing Nbr.       | 18.042-MNA-305.03         | Sh. 9         |
| Customer Doc. Nbr. |                           | Cont. 27      |

|      |                     |          |      |            |            |                                                         |                        |                    |                                                                                                             |          |                           |               |
|------|---------------------|----------|------|------------|------------|---------------------------------------------------------|------------------------|--------------------|-------------------------------------------------------------------------------------------------------------|----------|---------------------------|---------------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB                    | EPMV               | AUXILIARY MOTOR & RELAY<br>MVD-20 & MVD-21 20kV Boiler Plant<br><br>SCHEMATIC DIAGRAM<br><br>SBC-W OUTGOING | Scale    | NTS                       | = H03,H09,H10 |
|      |                     |          |      | Prepared   | YR         |                                                         |                        |                    |                                                                                                             | Customer | PT Multimas Nabati Asahan | & 305         |
| 00   | Issued for Approval | 11.01.19 | YR   | Checked    | AW         |                                                         | Drawing Nbr.           | 18.042-MNA-305.03  |                                                                                                             | Sh. 10   |                           |               |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on:                                               | BE318042               | Customer Doc. Nbr. |                                                                                                             |          | Cont. 27                  |               |
|      |                     |          |      |            |            |                                                         | PT. ABB Sakti Industri |                    |                                                                                                             |          |                           |               |

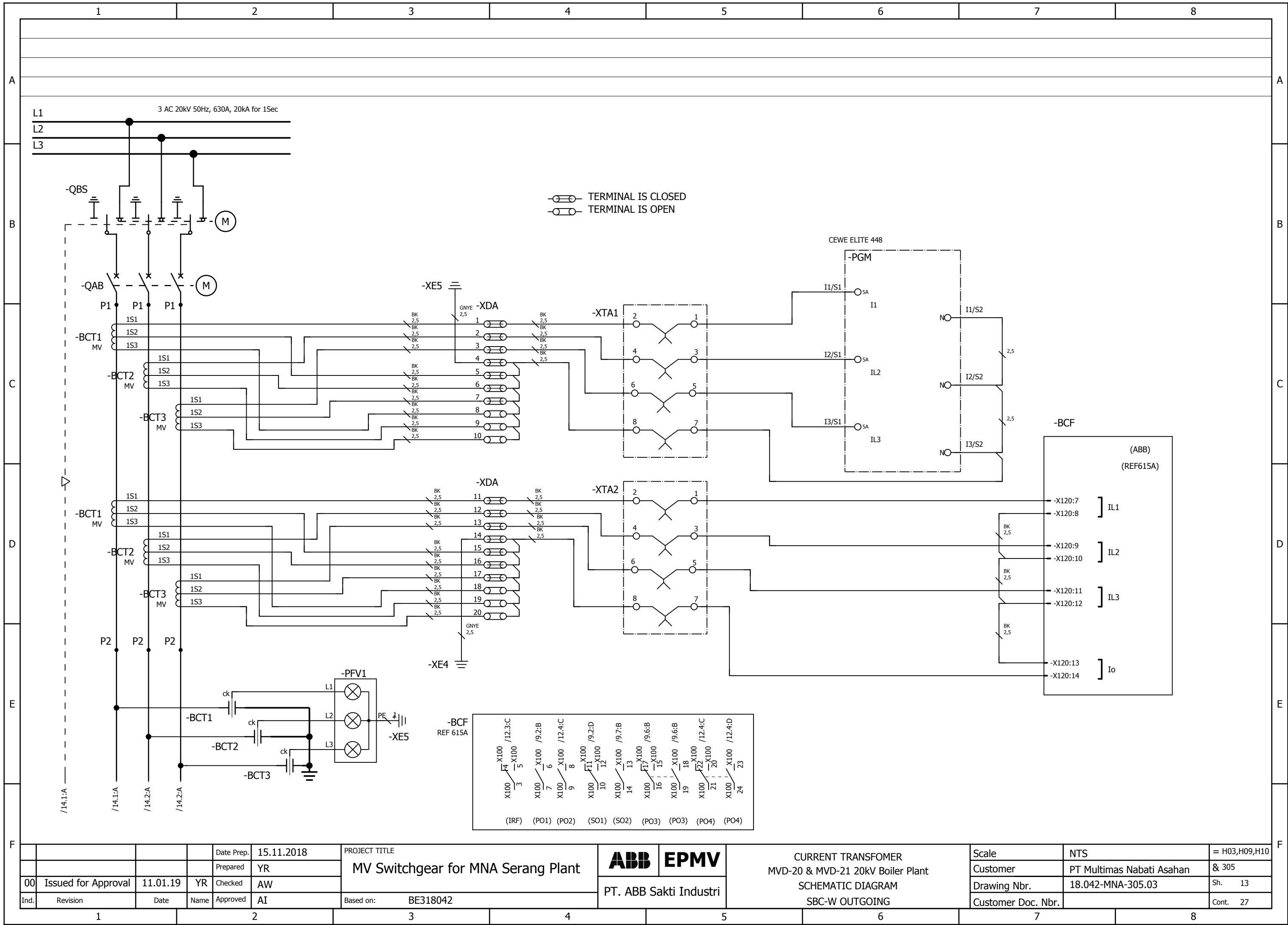
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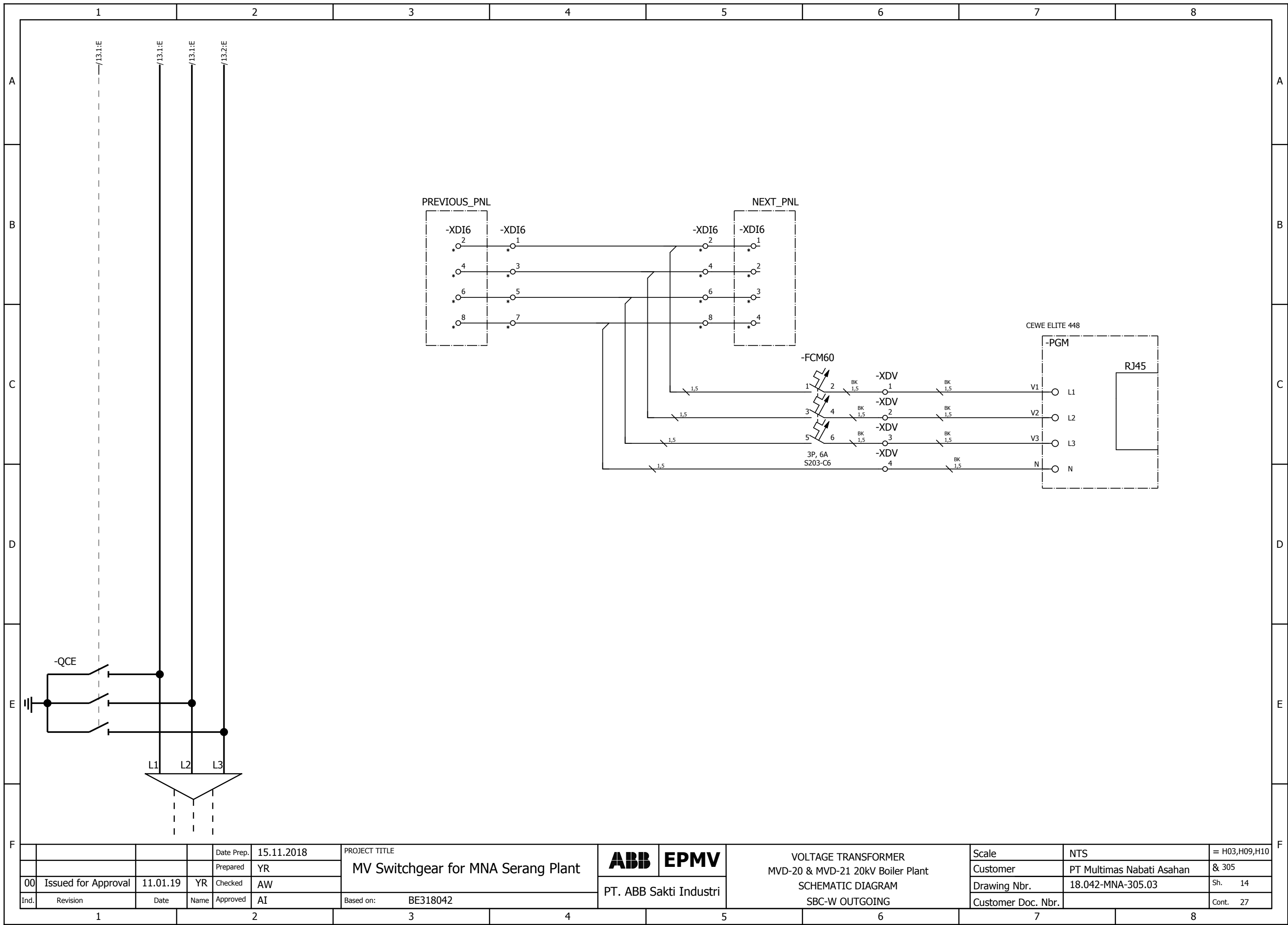


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|------|---------------------|----------|------|------------|------------|---------------------------------------------------------|------------------------|-------------------|-----------------------------------------------------------------------------------------------------------|--------------------|---------------------------|---------------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB                    | EPMV              | DIGITAL INPUT/OUPUT RELAY<br>MVD-20 & MVD-21 20kV Boiler Plant<br><br>SCHEMATIC DIAGRAM<br>SBC-W OUTGOING | Scale              | NTS                       | = H03,H09,H10 |
|      |                     |          |      | Prepared   | YR         |                                                         |                        |                   |                                                                                                           | Customer           | PT Multimas Nabati Asahan | & 305         |
| 00   | Issued for Approval | 11.01.19 | YR   | Checked    | AW         |                                                         | Drawing Nbr.           | 18.042-MNA-305.03 |                                                                                                           | Sh. 11             |                           |               |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on:                                               | BE318042               |                   |                                                                                                           | Customer Doc. Nbr. |                           | Cont. 27      |
|      |                     |          |      |            |            |                                                         | PT. ABB Sakti Industri |                   |                                                                                                           |                    |                           |               |

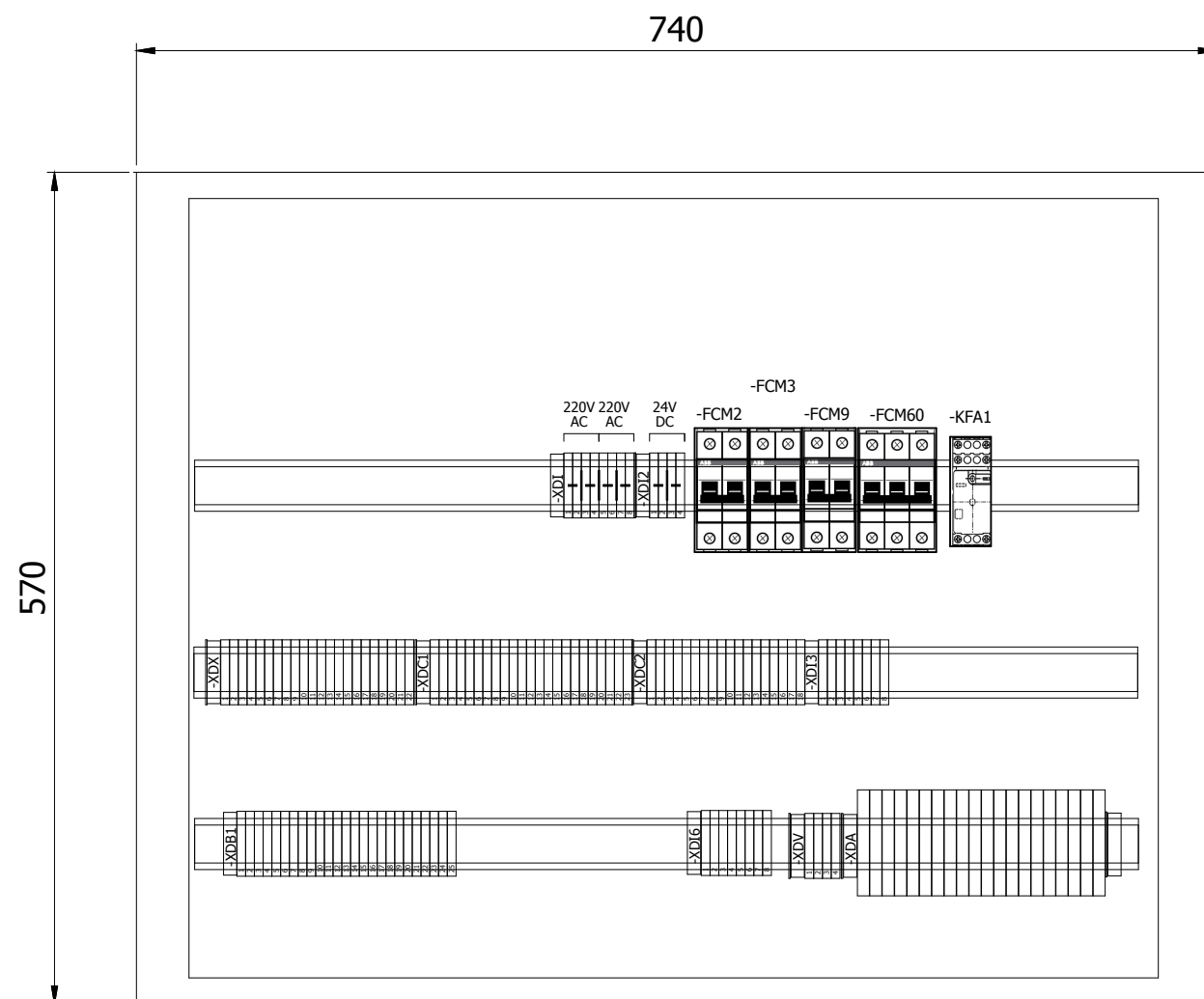


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| No. | EQUIPMENT DESCRIPTION                            | ENGRAVING                 |
|-----|--------------------------------------------------|---------------------------|
| 1   | -SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)     | -SFS, LOCAL-OFF-REMOTE    |
| 2   | -SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)          | -SFC, CB CLOSE            |
| 3   | -SFO (PUSH BUTTON CB OPEN, RED, ABB)             | -SFO, CB OPEN             |
| 4   | -PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB) | -PFY1, CB TRIP            |
| 5   | -PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)   | -PFR1, CB CLOSE           |
| 6   | -PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)  | -PFG1, CB OPEN            |
| 7   | -PGJ (DIGITAL MEASUREMENT, CEWE 448)             | -PGJ, DIGITAL MEASUREMENT |
| 8   | -BCF (FEEDER PROTECTION RELAY, REF615J, ABB)     | -BCF, RELAY PROTECTION    |
| 9   | -XTA1, XTA2 (TEST BLOCK , YONGSUNG)              | -XTA, CURRENT TEST BLOCK  |
|     |                                                  |                           |
|     |                                                  |                           |
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[illegible]

|      |                     |          |      |            |            |                                                         |                                                                                       |                                                                                       |                                                                                           |                   |                           |              |
|------|---------------------|----------|------|------------|------------|---------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------|---------------------------|--------------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant |  |  | LV BOX LAYOUT<br>MVD-20 & MVD-21 20kV Boiler Plant<br>SCHEMATIC DIAGRAM<br>SBC-W OUTGOING | Scale             | NTS                       | = H03,H09,H1 |
|      |                     |          |      | Prepared   | YR         |                                                         |                                                                                       |                                                                                       |                                                                                           | Customer          | PT Multimas Nabati Asahan | & 305        |
| 00   | Issued for Approval | 11.01.19 | YR   | Checked    | AW         |                                                         | PT. ABB Sakti Industri                                                                | Drawing Nbr.                                                                          |                                                                                           | 18.042-MNA-305.03 | Sh. 16                    |              |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on: BE318042                                      |                                                                                       | Customer Doc. Nbr.                                                                    |                                                                                           |                   | Cont. 27                  |              |

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|             | 1                     | 2                                                                                                               | 3                                         | 4 | 5                | 6                  | 7               | 8               |
| A           | Summarized parts list |                                                                                                                 |                                           |   |                  |                    |                 |                 |
|             | IDABB_F02_005-IDABB   |                                                                                                                 |                                           |   |                  |                    |                 |                 |
|             | Carry over: 0.00      |                                                                                                                 |                                           |   |                  |                    |                 |                 |
|             | TAG ID                | DESCRIPTION                                                                                                     |                                           |   | MANUFACTUR/BRAND | PART NUMBER        | QUANTITY        | REMARKS         |
|             | -BCF                  | REF615A HBFAAAAACBC1ANA11G<br>Configuration : A ; Power Supply : 48-250 Vdc; 100-240 Vac                        |                                           |   | ABB              | HBFAAAAACBC1ANA11G | 1               | /8.2:C          |
|             | -EB                   | HEATER<br>230VAC, 50W                                                                                           |                                           |   | SANYANG          | CDXN010033P0009    | 1               | /7.6:E          |
|             | -FCM2                 | MCB S202-C10<br>MCB. 2P, 10A                                                                                    |                                           |   | ABB              | 2CDS252001R0104    | 1               | /7.2:C          |
|             | -FCM3;-FCM9           | MCB S202-C6<br>MCB. 2P, 6A                                                                                      |                                           |   | ABB              | 2CDS252001R0064    | 2               | /7.5:C;/7.6:C   |
|             | -FCM60                | MCB S203-C6<br>MCB. 3P, 6A                                                                                      |                                           |   | ABB              | 2CDS253001R0064    | 1               | /14.6:C         |
|             | -KFA1                 | Auxiliary Relay<br>230AC 4COS                                                                                   |                                           |   | PHOENIX          | 1YRA3DL266230001   | 1               | /9.7:E          |
|             | -PFG1                 | PILOT LIGHT GREEN #CL-523G#230VAC<br>CL-523G                                                                    |                                           |   | ABB              | 1SFA619402R5232    | 1               | /8.4:E          |
|             | -PFR1                 | PILOT LIGHT RED #CL-523R#230VAC<br>CL-523R                                                                      |                                           |   | ABB              | 1SFA619402R5231    | 1               | /8.4:E          |
|             | -PFY1                 | PILOT LIGHT YELLOW #CL-523Y#230VAC<br>CL-523Y                                                                   |                                           |   | ABB              | 1SFA619402R5233    | 1               | /8.5:E          |
|             | -PGM1                 | POWER METER CEWE 448<br>Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC     |                                           |   | CEWE             | 1YRA4EA0350001     | 1               | /8.3:C          |
|             | B                     | -SFC                                                                                                            | PUSH BUTTON, YW1B-M1E10-G<br>GREEN COLOUR |   |                  | IDEC               | 1YRA3DA14700001 | 1               |
| -SFO        |                       | PUSH BUTTON, YW1B-M1E10-R<br>RED COLOUR                                                                         |                                           |   | IDEC             | 1YRA3DA14700002    | 1               | /9.4:B          |
| -SFS        |                       | LOCAL/REMOTE Selector Switch<br>Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position) |                                           |   | K&N              | 1YRA3DA17940011    | 1               | /9.2:B          |
| -TFS        |                       | Isolated PT100 Converter<br>Model : CN-6100-C1                                                                  |                                           |   | AUTONICS         | CN-6100-C1         | 1               | /8.7:C          |
| -XTA1;-XTA2 |                       | Current Test Block<br>Rated Current : 10A, 250V; Mounting : Flush Mounting                                      |                                           |   | YONGSUNG         | YSCTT-04C          | 2               | /13.4:C;/13.4:D |
| C           |                       |                                                                                                                 |                                           |   |                  |                    |                 |                 |
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| D           |                       |                                                                                                                 |                                           |   |                  |                    |                 |                 |
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| E           |                       |                                                                                                                 |                                           |   |                  |                    |                 |                 |
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| F           |                       |                                                                                                                 |                                           |   |                  |                    |                 |                 |
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